

5	(a) charges in metal do not move no (resultant) force on charges so no (electric) field (allow 1/2 for “no field inside sphere”)	B1 B1	[2]
	(b) <i>either</i> average field strength = $\frac{1}{2} (28 + 54) \text{ NC}^{-1}$	C1	
	average force = $8.5 \times 10^{-9} \times \frac{1}{2} (28 + 54)$ = $3.49 \times 10^{-7} \text{ N}$	C1	
	change in potential energy = $3.49 \times 10^{-7} \times 2.0 \times 10^{-2}$ = $7.0 \times 10^{-9} \text{ J}$ (allow 1 s.f.)	A1	
	(allow range 54 ± 1)		
	or (for a point charge) $V = Ex$	(C1)	
	$\Delta V = (54 \times 5.0 \times 10^{-2}) - (28 \times 7.0 \times 10^{-2})$	(C1)	
	change in potential energy = $8.5 \times 10^{-9} \times (2.70 - 1.96)$ = $6.3 \times 10^{-9} \text{ J}$ (allow 1 s.f.)	(A1)	
	(allow range 54 ± 1)		
	or ΔV is area under curve	(C1)	
	$\Delta V = 0.74 \text{ V}$	(C1)	
	change in potential energy = $8.5 \times 10^{-9} \times 0.74$ = $6.3 \times 10^{-9} \text{ J}$ (allow 1 s.f.)	(A1)	[3]
	(allow range 0.70 to 0.84)		